

NATHANIEL COREY

Electrical & Computer Engineering + Computer Science, Duke University

(443) 981-9231 | nathaniel.corey1@outlook.com | [linkedin.com/in/nathaniel-c-00b128255](https://www.linkedin.com/in/nathaniel-c-00b128255) | nathanielcorey.com

EDUCATION

Duke University — BSE Electrical & Computer Engineering, BS Computer Science Graduation: May 2027

GPA: 3.63 | Durham, NC

Relevant Coursework: Intro to Embedded Systems, Computer Architecture, Digital Systems, Signals and Systems, Microelectronic Devices and Circuits, Control Engineering, Operating Systems, Engineering Design and Communication, Intro to Artificial Intelligence

TECHNICAL SKILLS

Embedded & Systems: STM32 & TI Microcontrollers, I2C/UART/SPI, Motor Control Algorithms, Sensor Integration, Real-Time Systems, Hardware-in-the-Loop (HIL) Testing

Programming: C/C++, Python, Java, MATLAB/Simulink, Git, Automated Testing (pytest), Mobile App Development with Android SDK, Claude Code

Tools & Test Equipment: Oscilloscope, Logic Analyzer, JTAG/J-Link, Ozone Debugger, CCSTUDIO, SPICE, Wireshark, AutoCAD/Inventor

PROFESSIONAL EXPERIENCE

BETA Technologies — Airborne Software Engineering Intern May - Aug 2026

South Burlington, VT

- Designed and built a Hardware Development Kit (HDK) for verification & validation of airborne software on the TI AM263 MCU, used directly by a team of 5+ engineers
- Developed and validated drivers for I2C, UART, and SPI communication; debugged an I2C ACK/NACK fault using a logic analyzer and JTAG
- Built drivers via the MCU+ SDK to interface with FRAM and flash memory ICs; used a flash programming tool to configure compiled binaries and bootloader images
- Wrote a python API for an Analog Discovery 3 logic analyzer/waveform generator to capture I2C/UART/SPI protocol data and generate analog waveforms for test
- Designed a relay circuit for MCU/peripheral power control and built a GUI + API for remote HDK power control; used MATLAB/Simulink to design an RC low-pass filter for PWM-based analog waveform generation
- Wrote J-Link test scripts to set breakpoints at program symbols and resume execution, supporting early bring-up of the HDK's V&V environment

Stanley Black & Decker — Embedded Software Engineering Intern Jun - Aug 2025

Towson, MD

- Built a Python-based white-box testing framework (pytest, pandas) to validate STM32 motor controller firmware against safety and performance requirements
- Developed a Python/pandas module to derive motor speed, rotation direction, and fault conditions (invalid hall states, dropped/out-of-sequence transitions) from three-channel hall-effect sensor CSVs
- Profiled firmware performance using an Ozone debugger, oscilloscopes, and logic analyzer to identify and resolve system bottlenecks
- Led a cross-functional intern team applying data-driven analysis to improve manufacturing workforce retention strategies

RESEARCH

Independent Study, Interactive & Intelligent IoT Lab, Duke — Telemetry-Aware 5G Scheduling Fall 2026 - Present

- Characterized latency and synchronization behavior in a 5G-supported multi-user XR rehab system for Parkinson's patients, finding that an application-logic offloading feature increased network throughput by 24%.
- Developing a context-aware 5G RAN scheduler (C++) to prioritize latency-critical XR traffic, with a native telemetry agent embedded in an XR application process and a telemetry collector on the RAN

PROJECTS

Quadrotor Trajectory Control (MATLAB/Simulink) Sep - Dec 2025

- Designed and simulated a 6-DOF quadrotor controller using a nested PID architecture with feedforward acceleration, achieving >50% reduction in settling time and eliminating overshoot vs. baseline PID
- Validated robustness under a 20% mass mismatch, demonstrating integral action for disturbance rejection and steady-state error reduction

FPGA-Based Smart Solar Tracker May 2026

- Integrated photoresistor analog-to-digital conversion readings and servo motor control through an FPGA-based processor solution written in Verilog and MIPS assembly

Duke eNable — Leyman Prosthetic Arm Sep 2024 - Present

- Designed an EMG signal acquisition circuit using a differential amplifier and developed Arduino C programs for motor control and signal processing on a client-facing prosthetic arm